

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ 2 SET B

SUMMER SEMESTER, 2024-2025

DURATION: 30 MINUTES

FULL MARKS: 20

CSE 4205: Digital Logic Design

Programmable calculators are not allowed.

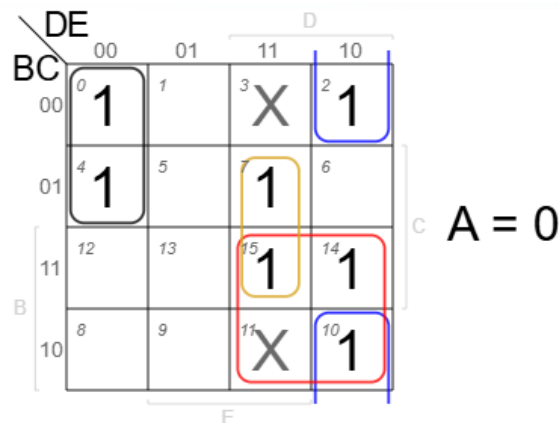
Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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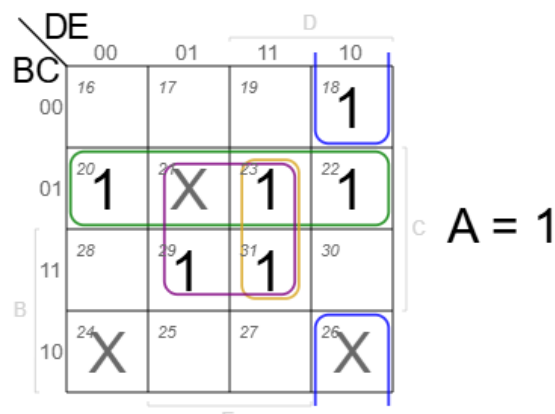
1. Use Karnaugh map to find the minimal form for the following expression (with don't care terms).
 Clearly group the terms contributing to your solution.

7
(CO3)
(PO2)

$$F(A, B, C, D, E) = \Sigma m(0, 2, 4, 7, 10, 14, 15, 18, 20, 22, 23, 29, 31), d(3, 11, 21, 24, 26)$$

Solution:

A = 0



A = 1

$$F = C'DE' + A'BD + AB'C + CDE + ACE + A'B'D'E'$$

Rubrics:

- 1 mark for correctly constructing the K-map and filling in the values (including don't-care terms).
- 0.5 mark (× 6) for correctly grouping the terms in the K-map.
- 0.5 mark (× 6) for correctly identifying the terms in the simplified expression (essential prime implicants).

2. Simplify the following expression using Quine McCluskey (Tabular Method). And in the process identify the Prime Implicants and Essential Prime Implicants.

13
(CO5)
(PO3)

$$F(A, B, C, D) = \sum m(2, 4, 7, 13, 15), d(1, 5, 12, 14)$$

Solution:

$$F(A, B, C, D) = \sum m(2, 4, 7, 13, 15), d(1, 5, 12, 14)$$

Step 0:

Group	Minterm	Binary Representation			
		A	B	C	D
1	m_1	0	0	0	1
1	m_2	0	0	1	0
1	m_4	0	1	0	0
2	m_5	0	1	0	1
2	m_{12}	1	1	0	0
3	m_7	0	1	1	1
3	m_{13}	1	1	0	1
3	m_{14}	1	1	1	0
4	m_{15}	1	1	1	1

Step 1:

Group	Minterm	Binary Representation				
		A	B	C	D	
1	m_1	0	0	0	1	✓
1	m_2	0	0	1	0	×
1	m_4	0	1	0	0	✓
2	m_5	0	1	0	1	✓
2	m_{12}	1	1	0	0	✓
3	m_7	0	1	1	1	✓
3	m_{13}	1	1	0	1	✓
3	m_{14}	1	1	1	0	✓
4	m_{15}	1	1	1	1	✓

Step 2:

Group	Matched Pairs	Binary Representation				
		A	B	C	D	
1	m_1, m_5	0	—	0	1	×
1	m_4, m_5	0	1	0	—	✓
1	m_4, m_{12}	—	1	0	0	✓
2	m_5, m_7	0	1	—	1	✓
2	m_5, m_{13}	—	1	0	1	✓
2	m_{12}, m_{13}	1	1	0	—	✓
2	m_{12}, m_{14}	1	1	—	0	✓
3	m_7, m_{15}	—	1	1	1	✓
3	m_{13}, m_{15}	1	1	—	1	✓
3	m_{14}, m_{15}	1	1	1	—	✓

	Group	Matched Pairs	Binary Representation				PI
			A	B	C	D	
Step 3:	0	m_2 (isolated)	0	0	1	0	$\overline{A}\overline{B}\overline{C}\overline{D}$
	1	m_1, m_5	0	—	0	1	$\overline{A}\overline{C}\overline{D}$
	2	m_4, m_5, m_{12}, m_{13}	—	1	0	—	$\overline{B}\overline{C}$
	2	m_5, m_7, m_{13}, m_{15}	—	1	—	1	$\overline{B}D$
	2	$m_{12}, m_{13}, m_{14}, m_{15}$	1	1	—	—	AB

	PI	Minterms	2	4	7	13	15	
Step 4:	$\overline{A}\overline{B}\overline{C}\overline{D}$	m_2	⊗					✓
	$\overline{A}\overline{C}\overline{D}$	m_1, m_5						
	$\overline{B}\overline{C}$	m_4, m_5, m_{12}, m_{13}		⊗		×		✓
	$\overline{B}D$	m_5, m_7, m_{13}, m_{15}			⊗	×	×	✓
	AB	$m_{12}, m_{13}, m_{14}, m_{15}$				×	×	

$$F = \overline{A}\overline{B}\overline{C}\overline{D} + \overline{B}\overline{C} + \overline{B}D$$

Rubrics:

- 1 mark for step 0.
- 3 marks for step 1.
- 3 marks for step 2.
- 3 marks for step 3. (Identifying prime implicants)
- 2 marks for step 4. (Identifying essential prime implicants)
- 1 mark for the final simplified expression.

Any correct simplification (with same number of terms and literals) of the expression will be accepted for both questions.